SECTIONS OF THE CONSTANT SHEAF ATTACHED TO A FLAT MODULE OVER A COHERENT RING ARE FLAT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file sites-modules.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites one related lemma, recorded in assembly.external_references with its canonical tag as a navigation pointer.

Allowed standard prerequisites (editorial).

- Flatness criterion via finitely generated ideals. Stacks Project, Algebra, Lemma algebra-lemma-flat (tag 00HD). Reduces flatness to injectivity of the maps induced by finitely generated ideals.
- Tensor products of constant sheaves with finitely presented modules. Stacks Project, Modules on Sites, Lemma lemma-tensor-with-finitely-presented (tag 09BK). Identifies the sheaf-level tensor product with the constant sheaf of the module tensor product.

Source reference. Original reference 00HD: sources/stacks-chow/tags.txt, source lines 639–639. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 093K: sources/stacks-chow/tags.txt, source lines 11518–11518. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $\mathcal{C}$ be a site. Let Λ be a coherent ring. Let M be a flat Λ-module. For $U \in \mathrm{Ob}(\mathcal{C})$ the module $\underline{M}(U)$ is a flat Λ-module.*

Proof. Let $I \subset \Lambda$ be a finitely generated ideal. By Algebra, Lemma 00HD it suffices to show that $\underline{M}(U) \otimes_\Lambda I \to \underline{M}(U)$ is injective. As Λ is coherent I is finitely presented as a Λ-module. By Lemma 093K we see that $\underline{M}(U) \otimes I = \underline{M \otimes I}$. Since M is flat the map $M \otimes I \to M$ is injective, whence $\underline{M \otimes I} \to \underline{M}$ is injective. □

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